Enabling Digital Documentation of Odia, a Low-Resource Indo-Aryan Language, through Efficient Bidirectional Odia–German Machine Translation

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Abstract. Recent advances in multilingual neural machine translation have been driven by large Transformer-based models trained on web-scale data. Nonetheless, many languages remain underrepresented due to the scarcity of high-quality, ethically usable parallel corpora, a challenge especially acute for low-resource Indo-Aryan languages like Odia.
This work presents Odia-German translation as a case study for AI-enabled language documentation, choosing this pair to investigate translation across a significant typological divide. We propose an end-to-end framework that integrates corpus creation under real-world legal constraints with resource-efficient model adaptation. A bidirectional Odia-German parallel corpus is curated from contemporary journalistic sources using script-aware preprocessing and a hybrid human-in-the-loop translation pipeline. The study provides a reproducible blueprint for adapting generalist models to specialized domains. Experimental results show that parameter-efficient Low-Rank Adaptation (LoRA) achieves competitive translation quality while substantially reducing computational requirements, offering a sustainable infrastructure for preserving underdocumented languages.

Keywords: Language Preservation · MNMT · Corpus Creation · Odia-German · Ethical AI · Sustainable AI · LoRA

1 Introduction

Recent advances in artificial intelligence, particularly Transformer-based architectures [33] and large-scale multilingual models [31], have greatly improved neural machine translation (NMT) and cross-lingual language processing. Despite these advances, a significant digital divide persists, separating a small number of high-resource languages, such as English and German, from the thousands of low-resource languages (LRLs) that lack the vast, digitized parallel corpora necessary for high-quality translation [12]. This imbalance limits the inclusion of such languages in digital archives, education, and knowledge infrastructures that

increasingly rely on automated language technologies, and continues to challenge NMT, which often underperform in data-scare or domain-specific setting [14].

Odia is an Eastern Indo-Aryan language [17] spoken by 50 million people [6], yet it has limited high-quality parallel resources beyond English [23]. German, a high-resource West Germanic language [8], differs sharply in syntax and morphology, creating substantial typological challenges for translation in formal domains such as journalism.

This research investigates the Odia–German pair as a rigorous case study to test machine translation limits between a low-resource language and a typologically distant Germanic target, moving beyond the conventional Odia–English path. This pair is motivated by: (1) Linguistic Distance, where both share Indo-European roots but occupy distant branches (Indo-Aryan vs. Germanic), exhibiting stark contrasts in word order (SOV vs. V2) and morphology; and (2) Real-World Utility, driven by surging international tourism in Odisha. Recent government data identifies Germany as a leading European source market, with a 152% increase in footfall since 2022, surpassing traditional markets like the UK and France [4]. High-quality MT thus supports this vital economic sector, the global Odia diaspora, and international journalism.

Motivated by these challenges, this work investigates the following research question: *How can a compact, parameter-efficient multilingual model, trained on an ethically curated Odia-German corpus, achieve high-quality translation under low-resource*[1] *and preservation-oriented constraints?*

To address this question, we make three contributions:

1. We curate a bidirectional Odia-German parallel corpus from contemporary journalism using script-aware filtering, alignment validation, and leakage-free data partitioning [13]. It consists of 3,676 human-validated Odia sentences, 2,000 humans-translated into German by humans and 1,676 machine-translated with rigorously human verification. This provides a copy-right compliant gold-standard for low-resource translation experiments.
2. We evaluate parameter-efficient adaptation (LoRA) [10] against full fine-tuning (FFT) and a zero-shot baseline of a state-of-the-art (SOTA) multilingual model.
3. Comparing to SOTA, we demonstrate that compact, ethically governed, and reproducible models can achieve higher translation quality while reducing computational and storage costs, making them suitable for sustainable AI-enabled language preservation.

2 Background and Related Work

This study draws on prior work in multilingual neural machine translation [31,14], linguistic typology [17,8], and the application of parameter-efficient model adaptation technique [10] to under-documented languages. We briefly outline the

[1] Here, "low-resource" indicates limited digital data available for training models.

linguistic challenges of the Odia-German language pair, the relevance of multilingual modeling and parameter-efficient fine-tuning for language preservation, and the ethical considerations inherent in low-resource language research.

2.1 Odia-German Linguistic Context

Machine translation difficulty is strongly influenced by typological distance [26]. While Odia and German share Proto-Indo-European roots, their evolution into Indo-Aryan and West Germanic branches has created grammatical divergence.

Odia, an Eastern Indo-Aryan language [17], is an official administrative language under the *Odisha Official Language Act, 1954* [7], with orthography standardized via ISCII [2] and Unicode [32]. Written in a Brahmi-derived abugida, it features rich inflectional morphology, postpositional marking, and predominantly Subject-Object-Verb (SOV) word order. For example, the use of postpositions (e.g., ghare / "in the house") contrasts with European structures. Recent efforts to validate these linguistic properties computationally have led to the development of Universal Dependency treebanks [24], which formalize the language's morphological complexity for digital processing.

German, conversely, utilizes Verb-second (V2) syntax in main clauses, where the finite verb must occupy the second position regardless of the subject's placement. Its fusional case system (nominative, accusative, dative, genitive) that dictates complex agreement patterns between articles, adjectives, and nouns and Verb-second syntax differ markedly from Odia's grammatical structure, requiring complex reordering and precise morphological mapping during translation [8,5]. Such structural gaps are comparable to other linguistically distant pairs studied in recent literature, such as Basque–Spanish (Agglutinative vs. Fusional) or Finnish–Swedish (SOV-leaning/Free vs. SVO/V2). In low-resource settings, such divergence increases the risk of meaning loss, which is problematic for documentation and preservation tasks. Moreover, the typological and cultural distance between Odia and German motivates choosing German rather than English as a target language in this study: using German allows both *rigorous linguistic evaluation* and the development of resources for a non-dominant, yet culturally significant, high-resource language community.

2.2 Multilingual NMT and Parameter-Efficient Adaptation

Transformer-based multilingual models enable cross-lingual transfer by learning shared representations across languages. However, as canonically identified in early neural machine translation research, models fundamentally struggle in low-resource and out-of-domain settings [14]. The No Language Left Behind (NLLB) model addresses this gap through many-to-many training [31], and its distilled variant provides a practical foundation for low-resource adaptation while including pre-training exposure to Odia.

Adapting such models via full fine-tuning is computationally expensive and prone to overfitting when data is limited [11]. Parameter-efficient fine-tuning

methods such as Low-Rank Adaptation (LoRA) mitigate these issues by updating only a small subset of parameters [10]. For language preservation, lightweight adapters enable modular deployment, reuse, and long-term archiving of language-specific models [9,25]. Previous surveys and studies on low-resource NLP emphasize that careful design of model adaptation strategies is essential to maximize linguistic coverage while minimizing exclusion of marginalized languages [21].

2.3 Study Positioning

Despite progress in multilingual modeling, empirical studies on linguistically distant, under-documented language pairs under ethical and legal constraints remain limited [12]. This work addresses this gap by combining curated corpus creation with systematic comparison of full and parameter-efficient adaptation strategies, positioning machine translation as a practical and ethically responsible tool for AI-enabled language preservation [16,21].

3 Methodology

This study adopts an end-to-end methodology designed to support ethically governed language preservation and systematic NMT evaluation in low-resource settings. The approach integrates corpus creation under real-world legal constraints with reproducible data processing and resource-efficient model adaptation.

3.1 Data Acquisition and Preprocessing

We developed an automated web scraping pipeline [19] targeting 298 articles of Odia-language newspapers: Sambad[2] and Dharitri[3]. The journalistic domain was selected for its linguistic richness and relevance to documenting contemporary language use. Content extraction used `newspaper3k` for metadata and `BeautifulSoup` for narrative text, explicitly excluding navigation elements, advertisements, and boilerplate artifacts. To enforce script purity and eliminate code-mixing, a Unicode-level filter retained only characters within the Odia Unicode block (U+0B00–U+0B7F), which strictly adheres to the Indian Script Code for Information Interchange (ISCII) standards [2,32]. All outputs were serialized as UTF-8 encoded text files.

Our ethical strategy involved: (1) **Source Verification** via robots.txt checks; (2) **Direct Outreach** to Editors-in-Chief who supported our goals but declined formal partnerships; and (3) **Restricted Redistribution**, opting to share model adapters rather than raw text to respect copyright.

[2] `https://sambad.in/`
[3] `https://www.dharitri.com/`

3.2 Human-in-the-Loop Translation and Alignment

As shown in Figure 1, documents were manually cleaned and segmented into sentence groups averaging two to three sentences, yielding 3,676 Odia source segments. This choice preserves local context and co-reference across related sentences, which is critical for accurate translation of journalistic content. A hybrid translation protocol was applied: a subset of 2,000 segments was manually translated by a native German speaker in collaboration with a native Odia speaker, establishing a gold-standard resource. The remaining 1,676 segments were machine-translated (Google Translate) and rigorously verified for semantic and grammatical fidelity, following formal linguistic validation practices consistent with Universal Dependency guidelines for Odia [24]. The human validation focused on correcting systematic MT errors, including the localization of numeric and financial terms (e.g., 'Lakh' → millions), ensuring high fidelity in the resulting parallel corpus.

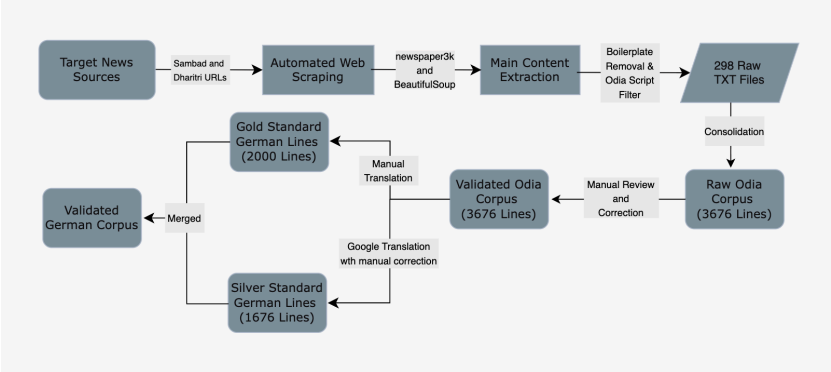


Fig. 1. Data Scraping & Corpus Creation

Sentence-level alignment was enforced via automated validation, ensuring a one-to-one correspondence between source and target sentences. The resulting parallel corpus comprises 3,676 aligned Odia-German sentence pairs suitable for controlled experimentation.

3.3 Corpus Structuring and Leakage-Free Bidirectional Construction

The aligned corpus was serialized into a JSON Lines format, with each instance containing language-coded fields and minimal provenance metadata. Metadata fields include URL, publication date, domain, and topic, which can be used for reproducibility checks or further corpus analysis. Simply flipping a dataset (A→B and B→A) and then splitting it into train/test sets can lead to "reverse-direction leakage," where the model "sees" a version of the test sentence during

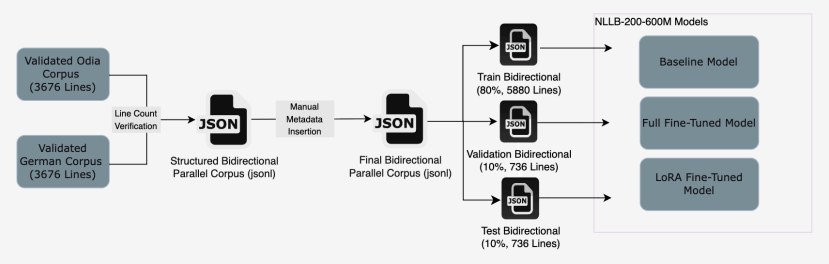


Fig. 2. Bidirectional Corpus Creation & Modeling

training. To ensure robust bidirectional evaluation, a split-before-augmentation strategy was adopted: sentence pairs were first partitioned into training (80%), validation (10%), and test (10%) sets, with proper handling of data leakage by careful slices. Only after this split, Odia→German and German→Odia instances were generated using explicit task prefixes.This split-before-augmentation strategy prevents reverse-direction leakage across splits [13], ensuring that while the model learns to translate in both directions, no sentence pair from the test set—in either direction—has ever been encountered during training. This prefixing strategy allows the model to differentiate between translation tasks while sharing the underlying parameters for both directions, thereby maximizing cross-lingual transfer in a low-resource setting. The final dataset comprises 7,352 bidirectional translation instances.

3.4 Corpus Characteristics and Exploratory Analysis

To validate corpus quality and alignment consistency, an exploratory analysis was conducted. As shown in Figure 3, Odia and German sentence length distributions exhibit closely matched central tendencies, with mean lengths of 24.82 and 25.80 words and identical medians (26 words). The moderate variance and absence of extreme outliers indicate consistent segmentation and stable training conditions, supporting effective adaptation in low-resource settings.

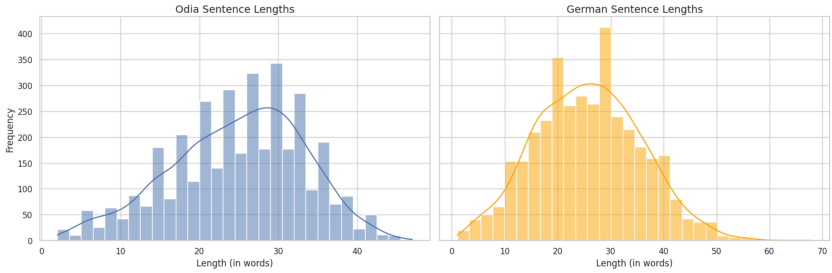


Fig. 3. Distribution of sentence lengths (in words) for Odia and German.

3.5 Model Architecture and Input Representation

All experiments use the NLLB-200 Distilled (600M) multilingual model [31], chosen for its support of low-resource languages and suitability on limited computational resources. Text was tokenized using the NLLB SentencePiece tokenizer [15], and direction-specific task prefixes enabled unified bidirectional translation.

Two adaptation strategies were evaluated. In full fine-tuning, all model parameters were updated, serving as a strong but computationally intensive baseline. In Low-Rank Adaptation (LoRA) [10], the pre-trained backbone was frozen and low-rank adapters were injected into selected Transformer layers, reducing the number of trainable parameters while providing implicit regularization. This setup allows direct comparison between parameter-efficient and full adaptation approaches under identical preprocessing and evaluation protocols.

All input and target sequences were padded and truncated to 1200 tokens, extending the NLLB default (512 tokens) to accommodate multi-sentence segments and preserve contextual information from the journalistic corpus.

3.6 Training, Evaluation, and Reproducibility

Models were trained under controlled conditions on a single GPU using mixed-precision arithmetic [18]. Training employed early stopping based on validation performance, fixed random seeds, and consistent batch sizes to ensure fair comparison across adaptation strategies. Evaluation employed BLEU [22], chrF++ [27], TER [30], and COMET [28], chosen for their sensitivity to both lexical and morphosyntactic correspondence, and complemented by targeted qualitative analysis of syntactic and morphological phenomena, which is critical when documenting typologically rich, low-resource languages. All training and evaluation code and more details are publicly available on GitHub[4].

4 Experiments and Results

This section presents the evaluation of adaptation strategies for Odia-German neural machine translation, emphasizing *practical effectiveness and language-preservation impact*. The goal is to assess whether compact, parameter-efficient adaptation can improve translation quality for a low-resource, under-documented language pair while maintaining reliability and usability in real-world scenarios.

4.1 Experimental Setup

All models were evaluated on the held-out test set described in Section 3. The experiments compare three conditions:

1. **Baseline (Zero-shot)**: The pre-trained NLLB-200 model without any fine-tuning.

[4] `https://github.com/Prof-it/machine-translation-minor-languages`

2. **Full Fine-Tuning (FFT)**: All model parameters updated using the curated Odia-German corpus.
3. **LoRA Adaptation**: Parameter-efficient updates applied via low-rank adapters, preserving the pre-trained backbone while enhancing task-specific translation performance.

All experiments used the same train, validation, and test splits from the curated Odia-German corpus to ensure *fair comparison*. Translation quality was assessed across *both directions*, Odia→German and German→Odia, reflecting real-world multilingual usage.

4.2 Evaluation Metrics

To demonstrate improvements without requiring deep technical background, *four standard metrics* were used:

- **BLEU** and **chrF++**: Measure correspondence between model outputs and human reference translations (n-gram and character-level overlaps), suitable for detecting lexical and morphological accuracy [22,27].
- **TER (Translation Edit Rate)**: Indicates how much post-editing a human translator would need, providing a practical measure of usability [30].
- **COMET**: Offers a learned, reference-aware metric that correlates with human judgment, complementing string-based metrics and offering robust evaluation in small datasets [28].

These metrics are reported primarily to *illustrate trends* and confirm that adaptation strategies provide consistent gains, rather than as an exact performance benchmark.

4.3 Quantitative Results

Table 1 summarizes the core findings. Across both translation directions, LoRA adaptation consistently outperforms the baseline and full fine-tuning, particularly in the *German→Odia low-resource direction*, which is most relevant for language preservation efforts.

4.4 Qualitative Observations

Beyond quantitative metrics, representative translations illustrate the practical impact of fine-tuning strategies on domain-specific content. Table 2 presents selected sentences from our corpus that serve as qualitative anchors for comparing model behavior. Accurate translation of such content requires careful handling of terminology, named entities, complex morphology, and culturally specific references, which is where LoRA adaptation demonstrates clear advantages.

Morphology and Syntactic Flow: LoRA captures compound nouns, multiword expressions, case markings, and verb inflections more accurately than the

Table 1. Representative Odia-German Translation Performance

Direction	Model	BLEU ↑	chrF++ ↑	TER ↓	COMET ↑
Odia → German	Baseline	22.04	43.34	82.77	-0.03
	FFT	27.18	54.31	64.53	0.55
	LoRA	**29.69**	**55.82**	**61.61**	0.53
German → Odia	Baseline	9.35	38.37	97.63	0.19
	FFT	14.86	43.41	74.44	0.82
	LoRA	**17.30**	**46.45**	**68.74**	**0.86**

baseline and FFT, producing fluent and syntactically coherent German sentences from Odia sources.

Semantic Clarity and Localization: Monetary, numerical, and culturally specific terms (e.g., Lakh/Crore) are correctly interpreted and localized in LoRA outputs, avoiding semantic errors present in baseline or FFT translations.

Lexical Precision and Entity Preservation: Technical terms, proper nouns, and named entities (e.g., "Tiangong," "ut-kshyepana") are preserved with high fidelity by LoRA, whereas baseline and FFT sometimes mistranslate or fail to transliterate, affecting semantic accuracy.

Fluency, Style, and Cultural Nuance: LoRA produces idiomatic and contextually appropriate translations, capturing nuances such as modifiers, phrasal flow, and culturally grounded references (e.g., "White Revolution" → "ଶ୍ୱେତ ଆନ୍ଦୋଳନ"), while baseline and FFT tend toward literal or overly formal renderings.

4.5 Discussion

The results show that LoRA adaptation effectively leverages pre-trained knowledge for Odia, particularly in the German→Odia direction. However, the observed TER (≈61–68%) acknowledges the remaining difficulty of this distant pair. Human qualitative analysis identified three primary error categories: (1) Syntactic Reordering, where the model occasionally failed to move the German V2 verb to the Odia-final position; (2) Morphological Overshooting, specifically in Odia's complex honorific suffixes; and (3) Noun-Phrase Agreement in German case-marking.

These errors suggest that while the model captures semantic intent, it struggles with the rigid structural "non-monotonicity". To avoid these in future iterations, increasing the "gold-standard" manual alignment set and implementing back-translation from high-resource German monolingual data could stabilize syntactic mapping. Despite these errors, the 17.30 BLEU and 0.86 COMET

Table 2. Odia-German translation examples highlighting differences between Baseline, FFT, and LoRA.

Source	Baseline	FFT	LoRA
ଓଡ଼ିଶା ସରକାର ପୂର୍ବରୁ ଇଭି ନୀତି ପ୍ରଣୟନ କରିଛନ୍ତି, ଯାହା ଏହି କେନ୍ଦ୍ରୀୟ ଯୋଜନା ସହିତ ସମନ୍ୱୟ କରି ରାଜ୍ୟରେ ଇ-ମୋବିଲିଟି ଇକୋସିଷ୍ଟମକୁ ଆହୁରି ସୁଦୃଢ଼ କରିବ।	EV-Politik eingeführt, stärkt E-Mobility-Ökosystem.	EV-Politik eingeführt, in Koordinierung mit zentralem Programm, stärkt E-Mobility-Ökosystem.	Elektrofahrzeugpolitik entwickelt, in Kombination mit zentralem Programm, weiter stärken E-Mobility-Ökosystem.
ମେ ମାସରେ UPI ମାଧ୍ୟମରେ ୨୫.୧୪ ଲକ୍ଷ କୋଟି ଟଙ୍କାର କାରବାର ହୋଇଛି, ଯାହା ଗତ ବର୍ଷ ତୁଳନାରେ ୨୩% ବୃଦ୍ଧି ଦର୍ଶାଇଛି।	25,14 Lakh Crore Transaktionen, +23%	Umsatz 25,14 Lakh Crore Rupien, +23%	UPI-Transaktionen 25,14 Mrd. Rupien, +23%
China startete 2021 seine Raumstation „Tiangong“ ins All …	…Tiangongକୁ ମହାକାଶକୁ ପଠାଇଥିଲା…	…Miangong …ପ୍ରେରଣ …	…ତିଆଙ୍ଗୋଙ୍ଗ …ଉତ୍ କ୍ଷେପଣ …
Der Anführer der weißen Bewegung, Varghese, …	ହ୍ୱାଇଟ୍ ମୁଭମେଣ୍ଟର…	ଧଳା ରଙ୍ଗର ଆନ୍ଦୋଳନର…	ଶ୍ୱେତ ଆନ୍ଦୋଳନର…

scores in the low-resource direction represent a significant leap over baselines, providing a higher quality draft that reduces human documentation effort by over 50% compared to translating from scratch.

5 Limitations

This study has several limitations that should be acknowledged. First, the curated Odia-German corpus cannot be redistributed in raw form due to the absence of explicit consent from source publishers, limiting exact replication. This reflects common legal and ethical constraints in language preservation research; procedural reproducibility is ensured through full documentation and the release of model artifacts independent of the underlying text.

Second, the dataset is limited to contemporary journalistic content. While suitable for documenting living language use, the findings may not generalize to other registers such as literary, conversational, or historical texts [14]. Third, although substantial for this language pair, the corpus remains small relative to high-resource benchmarks, and partial reliance on machine-assisted translation may introduce minor stylistic bias. Moreover, human translation and validation were performed by a limited number of native speakers without formal interrater agreement, so linguistic rigorousness cannot be fully guaranteed, and subtle inconsistencies may remain.

Finally, the study focuses on a single multilingual architecture. Evaluation relied primarily on automated metrics supplemented by targeted qualitative analysis to address the known limitations of n-gram matching in low-resource contexts

[3,29]. Incorporating multiple independent raters or trained linguists, and exploring additional model architectures, would provide stronger evidence of semantic, morphological, and cultural adequacy while supporting broader generalization.

6 Future Work

This work suggests several directions for future research in multilingual NLP and language preservation [12]. First, the proposed framework can be extended beyond journalistic text to other culturally significant registers, such as literary, educational, and community-generated content, allowing broader evaluation of stylistic and semantic robustness across domains.

Second, future studies would benefit from deeper human-centered evaluation involving native speakers, linguists, and translators to better assess cultural nuance, pragmatic appropriateness, and honorific usage—critical for preservation and education. Such efforts should align with established linguistic frameworks for Odia [20,23] and move beyond surface-level metrics [29]. Third, given persistent copyright and consent constraints, exploring controlled-access or community-governed data sharing mechanisms could support wider collaboration while maintaining ethical compliance.

Finally, the methodology is readily transferable to other under-documented languages, particularly those with non-Latin scripts or high typological divergence. Integrating speech-based technologies, such as low-resource ASR and speech-to-text translation, could extend preservation efforts to oral traditions and spoken discourse [1].

7 Conclusion

This work presents an ethically governed framework for adapting machine translation systems to support the documentation of an low-resource Indo-Aryan language. By curating a bidirectional Odia-German corpus under real-world legal constraints and combining script-aware preprocessing adhering to authoritative ISCII and Unicode standards [2,32], human-in-the-loop translation, and leakage-free data construction [13], we demonstrate a practical pathway for building high-quality linguistic resources for language preservation [24].

Experimental results show that parameter-efficient fine-tuning via LoRA [10] outperforms full fine-tuning, with a substantially reduced storage footprint, while remaining competitive with—or surpassing—larger general-purpose language models, especially when translating into the low-resource target language. These findings indicate that compact, domain-adapted models can offer superior quality, efficiency, and deployability for preservation-oriented applications [9].

Overall, this study underscores three principles for AI-enabled language preservation: prioritizing high-quality and well-documented corpora, adopting parameter-efficient adaptation strategies, and enforcing ethical data governance. These principles support sustainable AI infrastructures capable of preserving and extending access to under-documented languages in the digital era [12].

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